

Lecture 3 - The Impulse Response

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- Systems can be represented:
 1. As Difference Equations (DEQs).
 2. In State-Space Form.
 - The state-space form of a system is analogous to an infinite state machine

1 Represent the system $y(n) = x(n) + x(n-1) - y(n-1)$

1.I Find $\overrightarrow{S(n)}$

Let:

$$s_1(n) = x(n-1)$$

$$s_2(n) = y(n-1)$$

Then:

$$s_1(n+1) = x(n)$$

$$s_2(n+1) = y(n) = x(n) + x(n-1) - y(n-1) = x(n) + s_1(n) - s_2(n)$$

Thus, let $\overrightarrow{S(n)} = \begin{pmatrix} s_1(n) \\ s_2(n) \end{pmatrix}$.

1.II Find $\overrightarrow{S(n+1)}$

$$\begin{aligned} \overrightarrow{S(n+1)} &= \begin{pmatrix} s_1(n+1) \\ s_2(n+1) \end{pmatrix} \\ &= \begin{pmatrix} x(n) \\ x(n) + s_1(n) - s_2(n) \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} x(n) \end{aligned}$$

Therefore, our $\mathbf{A}$ and $\vec{B}$ matrices are $\begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ respectively.

1.III Find $y(n)$

$$\begin{aligned} y(n) &= x(n) + x(n-1) - y(n-1) \\ &= s_1(n) - s_2(n) + x(n) \\ &= \begin{pmatrix} -1 & 1 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n) \end{aligned}$$

Therefore, our $\mathbf{C}$ and $\mathbf{D}$ matrices are $\begin{pmatrix} -1 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \end{pmatrix}$ respectively.

Putting it all together, our system can be represented in state space form as:

$$\overrightarrow{S(n+1)} = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} x(n)$$

$$y(n) = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} x(n)$$

- The Kronecker-Delta function $\delta(n-k)$ is defined as:

$$- \delta(n-k) = \begin{cases} 1, & \text{if } n = k \\ 0 & \text{otherwise} \end{cases}$$

- The Kronecker-Delta function acts as the discrete analog to the continuous Dirac-Delta function.

2 Using the state space equations from example 1, compute the outputs to the system for the input $x(n)\delta(n-2)$

Current State ($\overrightarrow{S(n)}$)

Current Output ($y(n)$)

$$\overrightarrow{S(0)} = \vec{0}$$

$$y(0) = \begin{pmatrix} 1 & -1 \end{pmatrix} \overrightarrow{S(0)} + \begin{pmatrix} 1 \end{pmatrix} x(0) = 0$$

$$\overrightarrow{S(1)} = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix} \overrightarrow{S(0)} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} x(0) = \vec{0}$$

$$y(1) = \begin{pmatrix} 1 & -1 \end{pmatrix} \overrightarrow{S(1)} + \begin{pmatrix} 1 \end{pmatrix} x(1) = 0$$

$$\overrightarrow{S(2)} = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix} \overrightarrow{S(1)} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} x(1) = \vec{0}$$

$$y(2) = \begin{pmatrix} 1 & -1 \end{pmatrix} \overrightarrow{S(2)} + \begin{pmatrix} 1 \end{pmatrix} x(2) = 1$$

$$\overrightarrow{S(3)} = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix} \overrightarrow{S(2)} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} x(2) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$y(3) = \begin{pmatrix} 1 & -1 \end{pmatrix} \overrightarrow{S(3)} + \begin{pmatrix} 1 \end{pmatrix} x(3) = 0$$

$$\overrightarrow{S(4)} = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix} \overrightarrow{S(3)} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} x(3) = \vec{0}$$

$$y(4) = \begin{pmatrix} 1 & -1 \end{pmatrix} \overrightarrow{S(4)} + \begin{pmatrix} 1 \end{pmatrix} x(4) = 0$$

$\vdots$

$\vdots$

3 Model the system $y(n) = x(n) + \alpha y(n-1)$ in state-space form and compute its outputs for $x(n) = \delta(n)$.

Note that this system models compound interest.

3.a Express system in state-space form

3.a.I Find $S(n)$

Let:

$$s_1(n) - y(n-1)$$

Then:

$$s_1(n+1) = y(n) = x(n) + \alpha y(n-1) = \alpha s_1(n) + x(n)$$

Thus, let $\overrightarrow{S(n)} = (s_1(n))$

3.a.II Find $S(n+1)$

$$\begin{aligned}\overrightarrow{S(n+1)} &= \begin{pmatrix} s_1(n+1) \end{pmatrix} \\ &= \begin{pmatrix} \alpha s_1(n) + x(n) \end{pmatrix} \\ &= \begin{pmatrix} \alpha \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n)\end{aligned}$$

Therefore, our $\mathbf{A}$ and $\mathbf{B}$ matrices are $\begin{pmatrix} \alpha \end{pmatrix}$ and $\begin{pmatrix} 1 \end{pmatrix}$ respectively.

3.a.III Find $y(n)$

$$\begin{aligned}y(n) &= x(n) + \alpha y(n-1) \\ &= s_1(n) + x(n) \\ &= \begin{pmatrix} \alpha \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n)\end{aligned}$$

Therefore, our $\mathbf{C}$ and $\mathbf{D}$ matrices are $\begin{pmatrix} \alpha \end{pmatrix}$ and $\begin{pmatrix} 1 \end{pmatrix}$ respectively.

Putting it all together, our system can be represented in state space form as:

$$\begin{aligned}\overrightarrow{S(n+1)} &= \begin{pmatrix} \alpha \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n) \\ y(n) &= \begin{pmatrix} \alpha \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n)\end{aligned}$$

3.b Compute the inputs of $y(n)$

Current State ($\overrightarrow{S(n)}$)

Current Output ($y(n)$)

$$\overrightarrow{S(0)} = \vec{0}$$

$$y(0) = \begin{pmatrix} \alpha \end{pmatrix} \overrightarrow{S(0)} + \begin{pmatrix} 1 \end{pmatrix} x(0) = 1$$

$$\overrightarrow{S(1)} = \begin{pmatrix} \alpha \end{pmatrix} \overrightarrow{S(0)} + \begin{pmatrix} 1 \end{pmatrix} x(0) = 1$$

$$y(1) = \begin{pmatrix} \alpha \end{pmatrix} \overrightarrow{S(1)} + \begin{pmatrix} 1 \end{pmatrix} x(1) = \alpha$$

$$\overrightarrow{S(2)} = \begin{pmatrix} \alpha \end{pmatrix} \overrightarrow{S(1)} + \begin{pmatrix} 1 \end{pmatrix} x(1) = \alpha$$

$$y(2) = \begin{pmatrix} \alpha \end{pmatrix} \overrightarrow{S(2)} + \begin{pmatrix} 1 \end{pmatrix} x(2) = \alpha^2$$

$$\overrightarrow{S(3)} = \begin{pmatrix} \alpha \end{pmatrix} \overrightarrow{S(2)} + \begin{pmatrix} 1 \end{pmatrix} x(2) = \alpha^2$$

$$y(3) = \begin{pmatrix} \alpha \end{pmatrix} \overrightarrow{S(3)} + \begin{pmatrix} 1 \end{pmatrix} x(3) = \alpha^3$$

$\vdots$

$\vdots$

Generalizing, for input $x(n) = \delta(n)$, $y(n) = \alpha^n$ for this system.

4 Describe the outputs of any system for the input $x(n) = \delta(n - 1)$ using the state-space form of systems

We know that any system can be represented in state space form by the equations:

$$\begin{aligned}\overrightarrow{S(n+1)} &= \mathbf{A}\overrightarrow{S(n)} + \overrightarrow{B(n)}x(n) \\ y(n) &= \mathbf{C}\overrightarrow{S(n)} + \mathbf{D}x(n)\end{aligned}$$

For the input $x(n) = \delta(n)$, we can deduce that:

Current State ($\overrightarrow{S(n)}$)	Current Output ($y(n)$)
$\overrightarrow{S(0)} = \vec{0}$	$y(0) = \mathbf{C}\overrightarrow{S(0)} + \mathbf{D}x(0) = \mathbf{D}$
$\overrightarrow{S(1)} = \mathbf{A}\overrightarrow{S(0)} + \overrightarrow{B}x(0) = \overrightarrow{B}$	$y(1) = \mathbf{C}\overrightarrow{S(1)} + \mathbf{D}x(1) = \mathbf{C}\overrightarrow{B}$
$\overrightarrow{S(2)} = \mathbf{A}\overrightarrow{S(1)} + \overrightarrow{B}x(1) = \mathbf{A}\overrightarrow{B}$	$y(2) = \mathbf{C}\overrightarrow{S(2)} + \mathbf{D}x(2) = \mathbf{C}\mathbf{A}\overrightarrow{B}$
$\overrightarrow{S(3)} = \mathbf{A}\overrightarrow{S(2)} + \overrightarrow{B}x(2) = \mathbf{A}^2\overrightarrow{B}$	$y(3) = \mathbf{C}\overrightarrow{S(3)} + \mathbf{D}x(3) = \mathbf{C}\mathbf{A}^2\overrightarrow{B}$
$\vdots$	$\vdots$

Thus, we can describe the inputs to any system with input $x(n) = \delta(n)$ in state-space form as:

$$y(n) = \begin{cases} 0, & \text{if } n < 0 \\ \mathbf{D}, & \text{if } n = 0 \\ \mathbf{C}\mathbf{A}^{n-1}\overrightarrow{B}, & \text{if } n > 0 \end{cases}$$

- The impulse response of a system is defined as the outputs of a system when $x(n) = \delta(n)$
 - Denoted $h(n)$

5 Represent the system $y(n) = \alpha_1 x(n) + \alpha_2 x(n-1) + \alpha_3 x(n-2)$ in state-space form and describe its impulse response

Note that we can clearly take a shortcut here and realize that for $x(n) = \delta(n)$, $y(n) = \alpha_1 \delta(n) + \alpha_2 \delta(n-1) + \alpha_3 \delta(n-2)$, and thus:

$$h(n) = \begin{cases} \alpha_1, & \text{if } n = 0 \\ \alpha_2, & \text{if } n = 1 \\ \alpha_3, & \text{if } n = 2 \\ 0, & \text{otherwise} \end{cases}$$

However, we will do as the question asks

5.a Represent the system in state-space form

5.a.I Find $\overrightarrow{S(n)}$

Let:

$$s_1(n) = x(n-2)$$

$$s_2(n) = x(n-1)$$

Then:

$$s_1(n+1) = x(n-1) = s_2(n)$$

$$s_2(n+1) = x(n)$$

Thus, let $\overrightarrow{S(n)} = \begin{pmatrix} s_1(n) \\ s_2(n) \end{pmatrix}$.

5.a.II Find $S(n+1)$

$$\begin{aligned}\overrightarrow{S(n+1)} &= \begin{pmatrix} s_1(n+1) \\ s_2(n+1) \end{pmatrix} \\ &= \begin{pmatrix} s_2(n) \\ x(n) \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} x(n)\end{aligned}$$

Therefore, our $\mathbf{A}$ and $\overrightarrow{B}$ are $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ respectively.

5.a.III Find $y(n)$

$$\begin{aligned}y(n) &= \alpha_1 x(n) + \alpha_2 x(n-1) + \alpha_3 x(n-2) \\ &= \alpha_3 s_1(n) + \alpha_2 s_2(n) + \alpha_1 x(n) \\ &= \begin{pmatrix} \alpha_3 & \alpha_2 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} \alpha_1 \end{pmatrix} x(n)\end{aligned}$$

Therefore, our $\mathbf{C}$ and $\mathbf{D}$ matrices are $\begin{pmatrix} \alpha_3 & \alpha_2 \end{pmatrix}$ and $\begin{pmatrix} \alpha_1 \end{pmatrix}$ respectively.
Putting it all together, our system can be represented in state space form as:

$$\begin{aligned}\overrightarrow{S(n+1)} &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} x(n) \\ y(n) &= \begin{pmatrix} \alpha_3 & \alpha_2 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} \alpha_1 \end{pmatrix} x(n)\end{aligned}$$

5.b Compute $y(n)$ for $x(n) = \delta(n)$ using the state space equations

<u>Current State ($\vec{S}(n)$)</u>	<u>Current Output ($y(n)$)</u>
$\vec{S}(0) = \vec{0}$	$y(0) = \begin{pmatrix} \alpha_3 & \alpha_2 \end{pmatrix} \vec{S}(0) + \begin{pmatrix} \alpha_1 \end{pmatrix} x(0) = \alpha_1$
$\vec{S}(1) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \vec{S}(0) + \begin{pmatrix} 0 \\ 1 \end{pmatrix} x(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$	$y(1) = \begin{pmatrix} \alpha_3 & \alpha_2 \end{pmatrix} \vec{S}(1) + \begin{pmatrix} \alpha_1 \end{pmatrix} x(1) = \alpha_2$
$\vec{S}(2) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \vec{S}(1) + \begin{pmatrix} 0 \\ 1 \end{pmatrix} x(1) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$	$y(2) = \begin{pmatrix} \alpha_3 & \alpha_2 \end{pmatrix} \vec{S}(2) + \begin{pmatrix} \alpha_1 \end{pmatrix} x(2) = \alpha_3$
$\vec{S}(3) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \vec{S}(2) + \begin{pmatrix} 0 \\ 1 \end{pmatrix} x(2) = \vec{0}$	$y(3) = \begin{pmatrix} \alpha_3 & \alpha_2 \end{pmatrix} \vec{S}(3) + \begin{pmatrix} \alpha_1 \end{pmatrix} x(3) = 0$
$\vdots$	$\vdots$

Thus, the impulse response of the system is given by:

$$h(n) = \begin{cases} \alpha_1 & \text{if } n = 0 \\ \alpha_2 & \text{if } n = 1 \\ \alpha_3 & \text{if } n = 2 \\ 0 & \text{otherwise} \end{cases}$$

- A *Finite Impulse Response* (FIR) system is a system for where there exists some same after which the system's output is always equal to zero.
 - Expressed in mathematical notation, for an FIR system, $\exists N \in \mathbb{N}$ such that $\forall n \geq N, h(n) = 0$
 - FIR systems have a finite number of possible outputs for its impulse response.
 - FIR systems can be expressed as a linear combination of Kronecker-Delta functions.
- An *Infinite Impulse Response* (IIR) system is any system which is NOT an FIR system.
- All systems without feedback are FIR systems, but not all systems with feedback are IIR.